A-A-53468
25 May 1990

COMMERCIAL ITEM DESCRIPTION

FURNACE, DENTAL LABORATORY, ELECTRIC

The General Services Administration has authorized the use of this Commercial Item Description.

Salient Characteristics:

This commercial item description covers an electrically operated, heating plate type, burn-out and heat treating furnace, with pyrometer, automatic temperature controls and replacement heating plates, as specified below:

Type I. - Complete Furnace

Type II. - Replacement Heating Plates

Style 1 - Rear Plates

Style 2 - Side Plates

The furnace shall be a 115 volt, 50/60 Hz, single phase, electrically operated, burn-out and heat treating furnace with two replaceable rear and two replaceable side heating plates, digital readout pyrometer, automatic temperature control, and electrical cord suitable for use in dental laboratory operations. The furnace shall be equipped with a spring balanced door which opens downward. The door shall be equipped with two hinges and two door stops which hold the door in a horizontal position when fully opened. The door shall remain in the extreme open position to which it is moved, presenting a level shelf. The inner surface of the door shall have a raised section in the center to act as a work platform. The door shall be equipped with a safety interlock switch which disconnects both sides of the electrical line to the heating plates when the door is opened beyond two inches.

The furnace housing and legs shall be designed so that the furnace will properly stand on a flat surface. Each furnace shall be supplied with two glazed-surface ceramic trays, automatic temperature control, a ceramic vent tube, and a digital readout pyrometer.

AMSC N/A FSC 6520

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The digital pyrometer shall indicate the actual furnace temperature. The "set" temperature shall be indicated on the pyrometer and adjusted when the "Press-to-Set" switch is depressed. When the switch is released the pyrometer shall indicate the actual furnace temperature. The furnace shall be equipped with an automatic temperature control, which consists of: an 0-4 hour heat soak timer, with an electronic, digital status indicator showing either "Preset", "Holding" or "End" soak timer modes; an eight position heat rate control; and a "Heater on" indicator.

Housing. The furnace shall have two sections, an upper section and a lower section (instrumentation). The upper section shall be fabricated from 20 gauge (US), type 304 stainless steel. A removable upper rear panel shall be fabricated from 16 gauge (B&S), type 3003-H14 aluminum alloy. The lower section shall be fabricated from 18 gauge (US) cold rolled steel.

Door and door frame. The door and door frame shall be fabricated from 20 gauge (US), type 304 stainless steel.

Heating element. The heating element wire shall be fabricated from an alloy consisting of 5.5 percent aluminum (nominal), 22 percent chromium (nominal) and the balance of iron. The heating element shall be 18 gauge resistance wire.

Refractories and insulation.

Heating chamber. The insulation of the heating chamber shall be fabricated from alumina-silica, ceramic fiber. The blanket insulation used in addition to the ceramic fiber in the heating chamber shall be fabricated from an alumina-silica fiber blanket.

Refractory parts. The heating element supports, and the terminal block which is mounted on the rear insulating panel, shall be fabricated from a castable ceramic material.

Door insulation. The door insulation shall be fabricated from alumina-silica, ceramic fiber.

Insulation between housings. The upper and lower housings of the furnace shall be insulated by an air space of 5/8 inch. This distance shall be maintained by two metal spacer bars between the two sections.

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Ceramic trays. The trays shall be fabricated from a castable ceramic material to prevent the absorption of wax. Each tray shall be 3/8 inch high and shall have dimensions of 5-1/8 by 4-1/2 inches. The tolerances on the dimensions shall be ± 1/8 inch. The two trays shall fit on the floor of the heating chamber side by side with the door closed.

Automatic temperature control. The furnace shall be equipped with a completely automatic, built-in temperature control, by which the desired temperatures, heat rates (2), and hold times may be preset upon starting the furnace. The preset temperature shall be reached with power input to the heating elements determined by the microprocessor heat rate control. The desired temperature shall be maintained regardless of normal line voltage fluctuations and changes in the heating element resistance.

The automatic temperature control system shall be rated at least 125 percent of the maximum current drawn at 115 volts. Any temperature within the range of the furnace shall be accurately set and maintained within ± 10 degrees F (6 degrees C).

A heat soak timer shall be settable from 0-4 hours; there shall be a tone to indicate to the operator when the preset temperature time has elapsed. A digital display shall indicate the status of the soak timer, once the timer has been preset and is in operation. Operation of the tone shall be selected by a switch located on the front panel of the furnace. The position of the tone switch (on or off) shall not effect the operation of the soak timer or digital status display. Furnace shall be equipped with self diagnostic; indicator lights to identify cause for furnace malfunction.

Pyrometer. The furnace shall be equipped with a pyrometer for indicating the furnace temperature. The pyrometer shall be calibrated, as specified herein, over a minimum range of 0°F to 1999°F, in increments of 1°F. The pyrometer shall be of the digital readout type.

Thermocouple. The furnace shall be equipped with a 14 gauge (B&S) diameter, Chromel/Alumel (type K) thermocouple. The thermocouple shall extend 1-5/8 inches into the furnace heating chamber.

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Thermocouple failure protection. The furnace shall be equipped with a thermocouple failure protection circuit which shall prevent the furnace from heating in the event of thermocouple failure.

Electrical cord. Each furnace shall be provided with an electrical cord, no less than 6 feet in usable length. The cord shall be 3-wire, No. 14 gauge, a type SPT-3 thermoplastic insulated cord; or type HPN, HSJ or HSJO rubber jacketed cord. A 3 prong vinyl plug (NEMA Type 5-20P) shall be attached to one end and shall be suitable for use with receptacles specified in American National Standard Institute (ANSI) standard C73.11 for caps and receptacles; 125 volts, 16 amperes, 2 pole, 3-wire grounding type. A strain relief shall be provided where the power cord enters the lower furnace housing. The terminal connections inside the furnace shall show no evidence of strain when tested as specified herein.

The ground lead shall be so connected that all exposed metal parts of the furnace are electrically grounded. The furnace shall conform to the requirement of Underwriters Laboratories (UL), Inc. standards.

Removable components. The removable heating plates, both when furnished as part of type I and as separate replacement plates (type II), shall be fabricated from a castable ceramic material. There shall be four heating plates within the heating chamber, two of which are located at the rear of the chamber (style 1), and one on each side of the heating chamber (style 2). The heating plates shall be of the exposed-element type. The two rear heating elements shall be physically reversed so that the hard surface of the plates faces the inside of the heating chamber; the heating element wire in the two side heating plates shall remain exposed.

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Dimensions.

Heating chamber. The heating chamber shall have the following dimensions:

Width: 9-1/4 ± 3/8 inches
Height: 5 ± 3/8 inches
Working Depth: 5-3/8 ± 3/8 inches

Overall dimensions. The overall dimensions of the furnace including the door handle, with the door closed, shall not exceed:

Width: 14-5/8 inches
Height: 19 inches
Depth: 14 inches

Finish. The furnace housing, door and door frame shall have a 2B finish. All outside metal surfaces, excluding the control panel, shall be painted with a durable, high-heat resistant, enamel using commercial paint processes. The color shall be black, matte-finish enamel and may have an orange accent panel on the door. The control panel shall have a brushed aluminum finish, coated with clear epoxy, with symbolization print in black and baked-on.

Performance.

Calibration. The pyrometer indicated temperature shall be accurate within ± 25°F (14°C) at 1300°F (704°C) when tested as specified herein.

Heating rate. The furnace shall be connected to a regulated 115 volts, 60 Hertz, power supply, and the heat rate control set at maximum (position number 8), the time required to increase the temperature of the furnace from an ambient temperature 68°F (20°C) to a "set" temperature of 1600°F (871°C) shall not exceed 75 minutes.

Power rating. The maximum power consumed by the furnace shall be 1650 watts, when tested as specified herein.

Dielectric withstand. There shall be no electrical breakdown when tested as specified herein.

Leakage current. The leakage current shall be not more than 0.5 milliamperes when tested as specified herein.

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Identification marking. Each furnace shall have a name plate of corrosion-resisting metal, securely mounted in a conspicuous location, and permanently and legibly marked with the following information:

(a) Manufacturer's name or registered trademark
(b) Model number
(c) Voltage
(d) Frequency
(e) Power requirements

Service data. The contractor shall furnish, with each unit, two (2) copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts.

Workmanship. The furnace shall be free from defects which detract from their appearance, or impair their serviceability, or may endanger the safety of the operator. All metal components shall be free from burrs, sharp edges, and other deformities.

Unit. Each (EA). or Pair (PR). Each furnace, as specified, constitutes one unit; or one pair of side plates or one pair of rear plates, as specified, constitutes one unit.

Unit package. Each furnace shall be securely packaged to prevent breakage during shipping, handling and storage. Each pair of plates shall be securely packaged to prevent breakage during shipping, handling and storage.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Classification of inspection. The inspection of the furnace shall be classified as follows:

1. First article inspection
2. Quality conformance inspection

First article inspection. First article inspection shall consist of all examinations specified below and all testing specified herein and shall apply to the first article sample only. This inspection shall be performed only when required by the procuring activity.

Quality conformance inspection. Quality conformance inspection shall consist of all examinations specified and testing shall be as specified in tests and shall apply to those samples selected for inspection and testing. Quality conformance inspection shall be performed by the supplier as forementioned.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one furnace. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Chemical analysis. A sufficient quantity of each lot of metal used in the manufacture of the heating element wire shall be selected for chemical analysis in accordance with Federal Test Method Standard No. 151, Methods 111 or 112.

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Door counterbalance test. The counterbalance test shall be performed at the two extreme positions of the door (fully open and fully closed). With the door in the fully closed position, a spring balance of adequate capacity, graduated in two ounce intervals, shall be attached by a suitable means to the longitudinal center of the counterweight. A force shall be applied perpendicular to the longitudinal axis in the direction to open the furnace door. The force shall be slowly and continuously applied perpendicular to the longitudinal axis until the door is fully open. The maximum force required to completely open the door shall be recorded. The test as specified above shall be repeated in the opposite direction and the maximum force to fully close the door shall be recorded.

Strain relief. The electrical cord strain test shall be conducted in accordance with Underwriters Laboratories, Inc. Standard UL-1262.

Continuity to ground. A continuity indicating device shall be used to check continuity between the ground pin of the input plug at the end of the electrical cord and all exposed metal parts of the furnace. Grounding procedures shall be in accordance with UL-1262.

Calibration. Each furnace shall be tested for calibration accuracy by the use of Tempil pellets. A Tempil pellet, compounded to melt at 1300F (704°C) shall be used in accordance to the following procedure:

(a) With the furnace at, or near ambient temperature, place an empty, medium sized flask (inlay casting ring) approximately at the center of the heating chamber.

(b) Set a small, thin, metal plate or tray on top of the flask and place a 1300°F (704°C) Tempil pellet at the center of the tray.

(c) Close the furnace door and place the power switch in the "on" position.

(d) Set the heat rate control at position number 4, and set the desired temperature for 1000°F (538°).

(e) Once the pyrometer indicates a furnace temperature of 1000°F (538°C), allow the furnace to heat soak at this temperature for three minutes.

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(f) After the three minute heat soak period, set the desired temperature for 1500°F (816°C).

(g) When the pyrometer indicates a furnace temperature of 1200°F (649°C), begin to check for melting of the Tempil every 25°F (14°C). When checking, open the furnace door just enough to determine at a quick glance if the Tempil has begun to liquify around the edges. Keep the furnace door closed as much as possible during this observation period to prevent heat loss.

(h) When the Tempil pellet begins to melt or liquify around the edges, the pyrometer shall indicate 1300°F (704°C) ± 14°C). Tempil pellet has a melting point of 1300°F (704°C) ± 13°F) (±7°C).

Equilibrium temperature. The furnace shall be connected to a regulated 115 V, 60 H_z supply. The furnace shall be empty and the door closed. The temperature controller shall be set at 1700°F (926°C). Equilibrium shall be attained within 3 hours when test is performed at room temperature 68°F (20°C).

Power rating. The power consumed shall be determined by a suitable watt meter with the furnace operating at rated voltage and at the equilibrium temperature attained.

Dielectric strength test. All electrical component parts shall be tested after complete assembly and shall withstand without failure, a dielectric test of twice the rated voltage plus 1,000 volts or a total of 1,250 volts applied for one minute. This test shall be made on each of the individual lines of the circuit, using the metal frame as the other leg or ground of the 1,250 volt test.

Leakage current. The leakage current test shall be performed in accordance with UL-1262.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6520-01-089-5658	PLATE, HEATING, DENTAL LABORATORY FURNACE, Rear, Style 1, Type II.
6520-01-089-5659	PLATE, HEATING, DENTAL LABORATORY FURNACE, Side, Style 2, Type II.
6520-01-089-8282	FURNACE, DENTAL LABORATORY, ELECTRIC, Type I.

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Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

Project No. 6520-2057

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53468	FURNACE, DENTAL LABORATORY, ELECTRIC

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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